

# A Unified, Invariant, Geometry Based Measurement Theory for Machine Cognition: NCAP-10

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## Abstract

NCAP-10 is a management accessible measurement framework for evaluating how artificial and machine cognitive systems behave, adapt, and regulate themselves over time, independent of their underlying architecture. Unlike benchmark or human norm based evaluation, NCAP-10 represents a cognitive system as a geometric object on a difficulty manifold equipped with a family of measurement distributions and an induced Fisher information metric. From this object, NCAP-10 derives architecture invariant observables stability, sensitivity, synchrony, plasticity, memory retention, attractor structure, and capability as well as an economic layer that expresses value, risk, return on resources, and control cost as invariant functionals on the same geometric structure.

This revision closes the gaps present in the original NCAP-10 draft. Every abstract operator  $(\Phi, \Psi, \Omega, \mathcal{E})$  is given an explicit construction in terms of  $g_I$  and  $p(m | d)$ , illustrated by a fully worked example (a two channel difficulty manifold with a logistic measurement model). The admissible transformation group  $T$  is defined concretely as reparametrizations of the difficulty manifold together with sufficient statistic recodings of the observable space, and the distortion metric  $d_G$  on the space of information metrics is given explicitly as the affine invariant metric on symmetric positive definite matrices. The Invariance Theorem is proved directly from these constructions rather than asserted by definitional fiat. The Estimation Procedures section is completed and made operational for at least three concrete system classes (transformer based language models, recurrent/attractor networks, and neuromorphic spiking systems). Each section is accompanied by a diagram illustrating the corresponding structure.

## 1 Technical Preface

NCAP-10 represents a cognitive system as a tuple

$$N = (X, p(m | d), g_I, T),$$

where  $X \subseteq \mathcal{M}$  is an open subset of a smooth  $n$ -dimensional manifold  $\mathcal{M}$  (the *difficulty space*),  $p(m | d)$  is a smooth family of probability distributions over an observable space  $\mathcal{O}$  indexed by  $d \in X$ ,  $g_I$  is the Fisher information metric induced by this family, and  $T$  is a group of admissible transformations acting on  $(X, p)$  pairs.

Four operators extract invariant information from  $N$ :

$$\Phi : \mathcal{N}/\equiv \rightarrow \mathbb{R}^k, \quad \Psi : \mathcal{N}/\equiv \rightarrow \mathcal{D}, \quad \Omega : \mathcal{N}/\equiv \rightarrow \mathcal{I}, \quad \Xi : \mathcal{N}/\equiv \rightarrow \mathcal{F},$$

where  $\Phi$  returns a vector of *scalar* dynamical indices (stability, sensitivity, synchrony, plasticity all real numbers of the same type),  $\Psi$  returns a normalized difficulty embedding (an isometry class of  $(X, g_I)$  up to the action of  $T$ ),  $\Omega$  returns *discrete/topological* invariants (curvature spectra, Fisher volume, attractor count and homotopy type — cardinalities and topological types, deliberately kept separate from the real valued  $\Phi$ ), and  $\Xi$  returns *functional* invariants, i.e. elements of a function space  $\mathcal{F} = C([0, T], \mathbb{R}_{>0})$  (time indexed curves such as the memory retention curve). Keeping  $\Phi$ ,  $\Omega$ ,  $\Xi$  in three separate codomains ( $\mathbb{R}^k$ , discrete/topological types, function spaces) avoids conflating scalars, cardinalities, and functionals into a single composite object, as elaborated in Section 7.

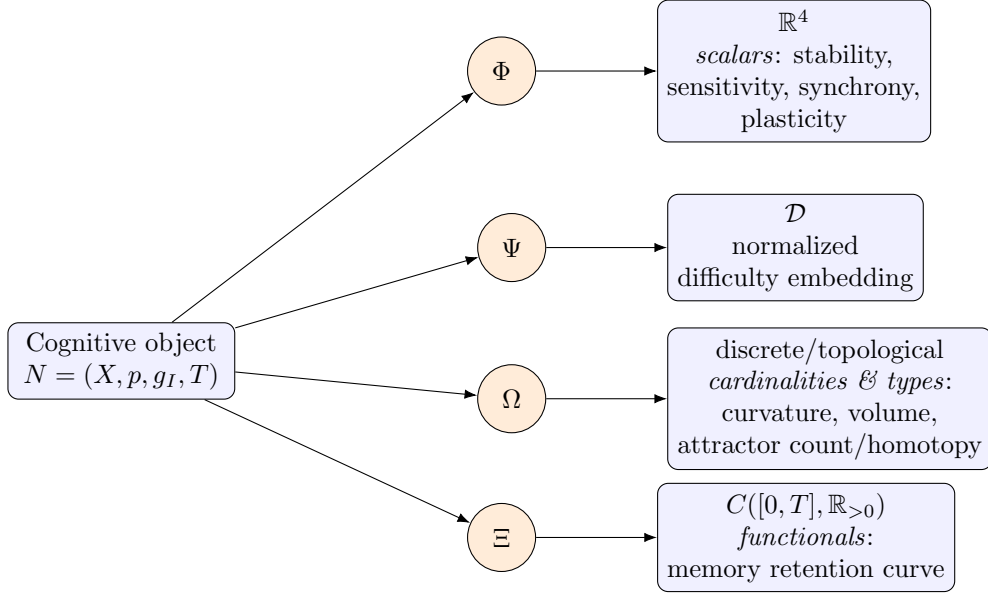


Figure 1: The NCAP-10 cognitive object and the four observable operators, with codomains kept type-separate:  $\Phi$  is purely real-valued scalars,  $\Omega$  is purely discrete/topological, and  $\Xi$  is purely functional (curves).

## 2 Introduction

Architecture specific benchmarks and human normed psychometrics cannot, in general, be applied across transformer based models, recurrent attractor networks, analog accelerators, and neuromorphic spiking processors on a common scale. NCAP-10 addresses this by treating cognition as a *dynamical process on a statistical manifold*: a system’s behavior under varying task difficulty  $d$  is represented by a conditional distribution  $p(m | d)$  over observables, and the local geometry of this distribution family captured by the Fisher information metric  $g_I$  — encodes sensitivity, stability, and regulatory structure.

The framework evaluates four organizational levels of a system:

- **Implementation level** — hardware substrate and resource allocation, entering through the cost functional  $C([N])$  (Section 9).
- **Learning level** — training and adaptation dynamics, entering through the admissible transformation group  $T$  acting on  $(X, p)$  (Section 6).
- **Memory level** — temporal evolution of  $g_I$  along a trajectory  $[N_t]$  (Sections 7 and 9).

- **Regulation level** — control actions realized as morphisms  $F \in T$  that minimize geometric distortion (Section 9).

Figure 2 shows how these four levels map onto the components of the cognitive object.

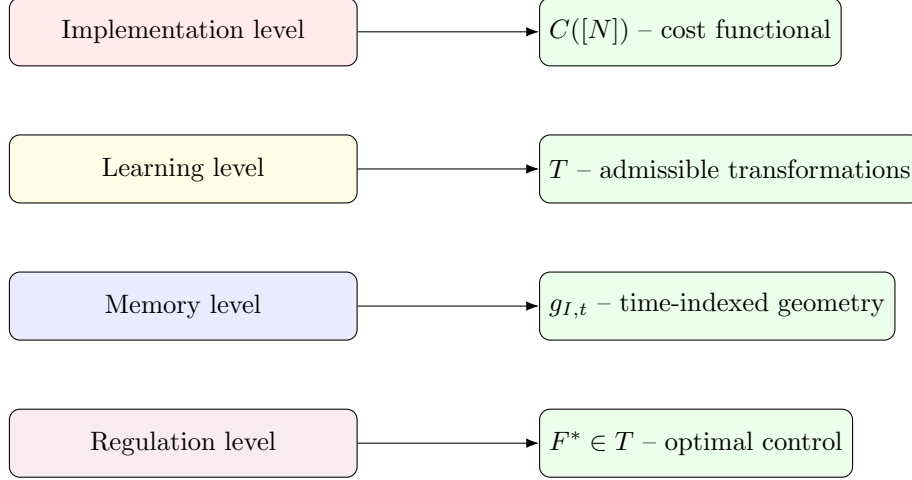


Figure 2: The four organizational levels addressed by NCAP-10 and the formal object representing each.

### 3 Primitive Object: The NCAP-10 Cognitive Object

**Definition 1** (Cognitive object). *A cognitive object is a tuple*

$$N = (X, p(m \mid d), g_I, T),$$

where:

- $X \subseteq \mathcal{M}$  is an open subset of a smooth  $n$ -manifold  $\mathcal{M}$ , with local coordinates  $d = (d^1, \dots, d^n)$  representing independently controllable difficulty dimensions (e.g. input length, distractor density, noise amplitude, distributional shift);
- $\mathcal{O}$  is a measurable space of observable outcomes, and for each  $d \in X$ ,  $p(\cdot \mid d) : \mathcal{O} \rightarrow [0, 1]$  is a probability density with respect to a fixed reference measure, jointly measurable in  $(m, d)$  and smooth in  $d$ ;
- $g_I$  is the Fisher information metric on  $X$  induced by the family  $\{p(\cdot \mid d)\}_{d \in X}$  (Definition 3), encoding the local sensitivity of the observable distribution to perturbations in difficulty;
- $T$  is the group of admissible transformations (Definition 5), representing the allowable re-descriptions of the system that preserve its cognitive identity.

Each component of  $N$  determines one axis of comparison across heterogeneous systems:

- $X$  specifies what difficulty dimensions are varied;
- $p(m \mid d)$  specifies how the system responds probabilistically;
- $g_I$  specifies how sensitively the response changes with difficulty;

- $T$  specifies which transformations preserve cognitive identity.

Together, these components define the NCAP-10 cognitive object as a coordinate free, geometry based representation of machine cognition, suitable for invariant comparison, risk analysis, and functorial economic interpretation.

## 4 Formal Definitions (Management Version)

**Difficulty space.** The set of all task or environmental conditions a system may face, represented as a smooth multi-dimensional space rather than a single difficulty score.

**Measurement behavior.** For each difficulty condition, the full distribution of possible outcomes not just an average describing how the system responds and how variable that response is.

**Information geometry.** A measure of how strongly a small change in difficulty changes the system’s response distribution. Large information-geometric “distance” for a small change in difficulty means the system is highly sensitive there; small distance means the system is locally stable.

**Admissible transformations.** Relabelings of the difficulty axes and recodings of the observable space (for example, switching from raw outputs to a sufficient statistic such as accuracy-vs-confidence) that do not change what the system actually does. Two systems related by such a relabeling are NCAP-10 equivalent.

**Observable operators.** Three outputs – a vector of dynamical indices ( $\Phi$ ), a normalized difficulty map ( $\Psi$ ), and a set of geometric invariants ( $\Omega$ ) – computed from  $g_I$  and  $p(m | d)$  so that relabeled systems produce identical outputs.

## 5 Formal Definitions (Technical Version)

**Definition 2** (Difficulty manifold).  $\mathcal{M}$  is a smooth  $n$ -manifold with atlas  $\mathcal{A}$ ; the difficulty space is an open connected  $X \subseteq \mathcal{M}$ .

**Definition 3** (Measurement model). A measurement model on  $X$  is a smooth map  $p : \mathcal{O} \times X \rightarrow [0, \infty)$  such that  $\int_{\mathcal{O}} p(m | d) dm = 1$  for all  $d \in X$ , and  $\partial_{d^i} \log p(m | d)$  exists and is square-integrable for every  $i$ .

**Definition 4** (Fisher information metric). The information metric  $g_I$  is the Riemannian metric on  $X$  with components

$$g_I(d)_{ij} = \mathbb{E}_{m \sim p(\cdot | d)} [\partial_{d^i} \log p(m | d) \partial_{d^j} \log p(m | d)].$$

By regularity,  $g_I(d)$  is symmetric positive semi-definite for every  $d$ ; we additionally require  $g_I(d) \succ 0$  (positive definite) on  $X$ , i.e. the measurement model is non-degenerate.

**Definition 5** (Admissible transformation group). Fix a reference measure  $\mu$  on  $\mathcal{O}$  (so that  $p(\cdot | d)$  denotes a density  $dP_d/d\mu$ ). An admissible transformation is a pair  $F = (\varphi, \psi)$  where  $\varphi : X \rightarrow X$  is a diffeomorphism and  $\psi : \mathcal{O} \rightarrow \mathcal{O}$  is a measurable bijection such that  $\psi_*\mu \ll \mu$  and  $\mu \ll \psi_*\mu$  (i.e.  $\psi$  is non-singular with respect to  $\mu$ , admitting a Radon–Nikodym derivative  $r_\psi := d(\psi_*\mu)/d\mu$ ,  $r_\psi > 0$   $\mu$ -a.e.).  $F$  acts on a cognitive object by

$$F \cdot N = (\varphi(X), p_F(m | d) := p(\psi^{-1}(m) | \varphi^{-1}(d)) \cdot r_{\psi^{-1}}(m), g_I^F, T), \quad r_{\psi^{-1}} := d(\psi_*^{-1}\mu)/d\mu,$$

where  $g_I^F$  is the Fisher metric induced by  $p_F$ ;  $r_{\psi^{-1}}$  is the Radon–Nikodym density correction making  $p_F(\cdot \mid d)$  integrate to 1 against  $\mu$  (this is the change-of-density formula for the pushforward  $\psi_*^{-1}P_{\varphi^{-1}(d)}$ ). Two important special cases:

- if  $\psi$  is measure-preserving ( $\psi_*\mu = \mu$ ), then  $r_{\psi^{-1}} \equiv 1$  and  $p_F(m \mid d) = p(\psi^{-1}(m) \mid \varphi^{-1}(d))$ , the pure relabeling case;
- if  $\mathcal{O}$  is a manifold and  $\mu$  is Lebesgue measure,  $r_{\psi^{-1}} = |\det J_{\psi^{-1}}|$ , recovering the Jacobian factor of the original formulation as a special case of the Radon–Nikodym derivative.

All statements below (Propositions 1–2, Theorem 1) hold for general non-singular  $\psi$ ; no additional measure-preservation hypothesis is required, because the reward and cost functionals  $R, C$  are required to be invariant under the induced action on  $\mathcal{O}$  (Definition 14), which is stated in  $\mu$ -density terms and so automatically absorbs  $r_{\psi^{-1}}$ .

The set of all pairs  $F = (\varphi, \psi)$  satisfying

$$\varphi^* g_I = g_I \quad (\text{exact information-metric preservation})$$

forms a group under composition, denoted  $T_0$ . For  $\varepsilon \geq 0$ , the  $\varepsilon$ -admissible group  $T_\varepsilon$  is the set of pairs  $F$  such that

$$d_G(\varphi^* g_I(d), g_I(d)) \leq \varepsilon \quad \text{for all } d \in X,$$

where  $d_G$  is the affine-invariant metric on the space of symmetric positive definite matrices (Definition 6). We write  $T := T_\varepsilon$  for a fixed tolerance  $\varepsilon \geq 0$  chosen at evaluation time;  $T_0 = T_0$  is the strict ( $\varepsilon = 0$ ) special case.

**Definition 6** (Affine-invariant metric on SPD matrices). For symmetric positive definite  $n \times n$  matrices  $A, B$ , define

$$d_G(A, B) = \left\| \log(A^{-1/2} B A^{-1/2}) \right\|_F = \left( \sum_{i=1}^n (\log \lambda_i)^2 \right)^{1/2},$$

where  $\lambda_1, \dots, \lambda_n$  are the eigenvalues of  $A^{-1/2} B A^{-1/2}$  and  $\|\cdot\|_F$  is the Frobenius norm.  $d_G$  is a genuine metric on the space of SPD matrices (the symmetric space  $GL(n)/O(n)$ ), invariant under  $A \mapsto P^\top A P$ ,  $B \mapsto P^\top B P$  for any invertible  $P$  [5].

**Definition 7** (Cognitive object space and equivalence).  $\mathcal{N}$  is the set of all cognitive objects satisfying Definitions 1–4. Two objects  $N, N' \in \mathcal{N}$  are NCAP-10 equivalent,  $N \equiv N'$ , if there exists  $F \in T$  with  $F \cdot N = N'$  (up to the obvious identification of  $(X, p)$  pairs related by  $F$ ). This is an equivalence relation because  $T$  is a group:  $\text{id} \in T$ ,  $F \in T \Rightarrow F^{-1} \in T$ , and  $T$  is closed under composition.

## 6 Worked Example: A Two-Channel Difficulty Manifold

To make every later construction concrete, fix the following minimal cognitive object, used as a running example throughout the rest of the paper.

**Example 1** (Logistic two-channel system). Let  $X = (0, \infty)^2 \subset \mathbb{R}^2$  with coordinates  $d = (d^1, d^2)$ , interpreted as two independent difficulty channels (e.g. input-length stress and distractor density).

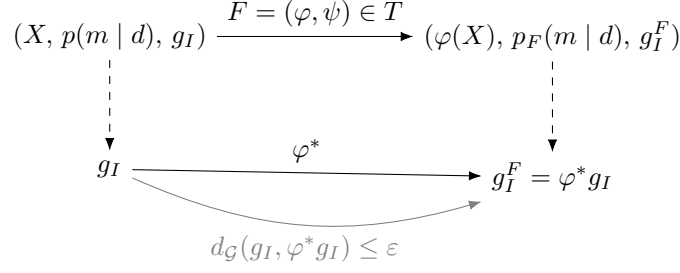


Figure 3: An admissible transformation  $F = (\varphi, \psi) \in T_\varepsilon$  relabels the difficulty manifold and the observable space while keeping the induced Fisher metric within distortion  $\varepsilon$  (measured by  $d_{\mathcal{G}}$ ) of the original.

Let  $\mathcal{O} = \{0, 1\}^2$  (two binary task outcomes, e.g. “primary task correct”, “secondary/monitoring task correct”). Define the measurement model by independent logistic responses,

$$p(m \mid d) = \prod_{i=1}^2 \sigma(-d^i)^{1-m_i} (1 - \sigma(-d^i))^{m_i}, \quad \sigma(x) = \frac{1}{1 + e^{-x}},$$

so that channel  $i$  succeeds ( $m_i = 1$ ) with probability  $1 - \sigma(-d^i) = \sigma(d^i)$ , decreasing toward chance as  $d^i$  grows. Each channel is an independent Bernoulli logistic model, so the Fisher information is diagonal:

$$g_I(d) = \begin{pmatrix} \sigma(d^1)(1 - \sigma(d^1)) & 0 \\ 0 & \sigma(d^2)(1 - \sigma(d^2)) \end{pmatrix}.$$

Fisher information for a single logistic channel (Example 1)

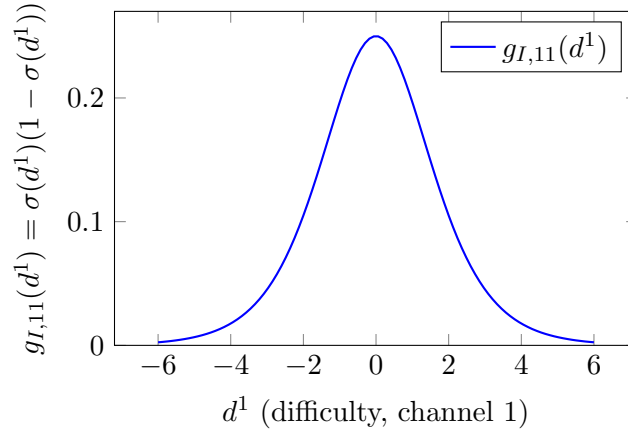


Figure 4: Each diagonal entry of  $g_I$  peaks at  $d^i = 0$  (maximum sensitivity, the system is maximally distinguishable from small perturbations) and decays toward 0 as  $|d^i| \rightarrow \infty$  (the system saturates and becomes locally insensitive — a stable but uninformative regime).

This example is used in Sections 7–9 to instantiate  $\Phi, \Psi, \Omega$ , the estimation procedures, and the economic functor.

## 7 Operational Implications: From Geometry to Cognitive Indices

NCAP-10 defines the dynamical indices named in the abstract directly as functionals of  $g_I$  and  $p(m \mid d)$ , organized by which entries of  $g_I$  they read.

**Definition 8** (Sensitivity index).

$$S(d) = \text{tr } g_I(d).$$

$S(d)$  aggregates the local distinguishability of the system's response across all difficulty channels at  $d$ .

**Definition 9** (Stability index).

$$\text{Stab}(d) = \frac{1}{1 + S(d)} \in (0, 1].$$

A system is locally stable at  $d$  if small perturbations of difficulty produce small changes in its response distribution, i.e.  $S(d)$  is small.

**Definition 10** (Synchrony index). For  $n \geq 2$  difficulty channels, define

$$\text{Sync}(d) = \frac{\sum_{i \neq j} |g_I(d)_{ij}|}{\sum_{i,j} |g_I(d)_{ij}|} \in [0, 1].$$

$\text{Sync}(d)$  measures the fraction of total Fisher information carried by cross-channel (off-diagonal) terms, i.e. the degree to which the system's response to one difficulty dimension is statistically coupled to another.

**Definition 11** (Plasticity functional). Let  $\{N_\tau\}_{\tau \in [0,1]}$  be a one-parameter family of cognitive objects representing a system before ( $\tau = 0$ ) and after ( $\tau = 1$ ) an adaptation step (e.g. a gradient update). Define

$$\text{Plast} = d_{\mathcal{G}}(g_{I,0}(d_0), g_{I,1}(d_0))$$

at a fixed reference difficulty  $d_0$ , using the metric of Definition 6.  $\text{Plast}$  measures how much the local information geometry changed due to adaptation.

**Definition 12** (Attractor set — a component of  $\Omega$ ). A point  $d^* \in X$  is an attractor of  $T$ -action restricted to a one-parameter subgroup  $\{F_t\} \subset T$  (representing the system's autonomous dynamics) if  $\varphi_t(d) \rightarrow d^*$  as  $t \rightarrow \infty$  for  $d$  in a neighbourhood of  $d^*$ . The attractor set  $\mathcal{A}(N) \subseteq X$  is the union of all such points. Its cardinality  $|\mathcal{A}(N)| \in \mathbb{N} \cup \{\infty\}$  and its homotopy type are discrete/topological data and are reported as part of  $\Omega([N])$ , not as a coordinate of  $\Phi([N])$ : cardinalities and topological types are a different kind of mathematical object from the real-valued indices  $S, \text{Stab}, \text{Sync}, \text{Plast}$  and must not be packed into the same vector space.

**Definition 13** (Memory-retention functional — the value of  $\Xi$ ). For a time-indexed family  $[N_t]_{t \in [0,T]}$ , define  $\rho(t) = -\frac{d}{dt} \log \sqrt{\det g_I(d^*, t)}$  at a fixed reference  $d^*$ , and

$$\xi(t) := \exp\left(-\int_0^t \rho(s) ds\right) = \frac{\sqrt{\det g_I(d^*, t)}}{\sqrt{\det g_I(d^*, 0)}} \in \mathbb{R}_{>0}.$$

$\Xi([N]) := \xi(\cdot) \in C([0, T], \mathbb{R}_{>0})$  is the entire retention curve, a functional invariant. The scalar  $V_{\text{mem}}([N]) := \xi(T)$  used in Section 9 is the evaluation functional  $\text{ev}_T : \mathcal{F} \rightarrow \mathbb{R}_{>0}$ ,  $\text{ev}_T(\xi) = \xi(T)$ , applied to  $\Xi([N])$  — i.e.  $V_{\text{mem}} = \text{ev}_T \circ \Xi$  is explicitly a composite of a functional ( $\Xi$ ) and a type-lowering evaluation map ( $\text{ev}_T$ ), not an independent coordinate of  $\Phi$ .

These four quantities —  $S$ ,  $\text{Stab}$ ,  $\text{Sync}$ ,  $\text{Plast}$  — are *all and only* the coordinates of  $\Phi([N]) \in \mathbb{R}^4$ : every coordinate of  $\Phi$  is a real-valued, dimensionless functional of  $g_I$  (and its first variation, for  $\text{Plast}$ ) evaluated at a point or pair of points, so  $\Phi$  is type-pure.  $\Omega([N])$  separately collects  $(|\mathcal{A}(N)|, \text{HomotopyType}(\mathcal{A}(N)), \text{Ric-spectrum}, \text{Vol}_{g_I}(X))$  — topological/spectral data — and  $\Xi([N]) = \xi(\cdot)$  is the memory-retention curve in  $C([0, T], \mathbb{R}_{>0})$ .  $\Psi([N])$  is the isometry class  $(X, g_I)/T$ , represented canonically by the normal coordinates of  $g_I$  at its volume centroid.

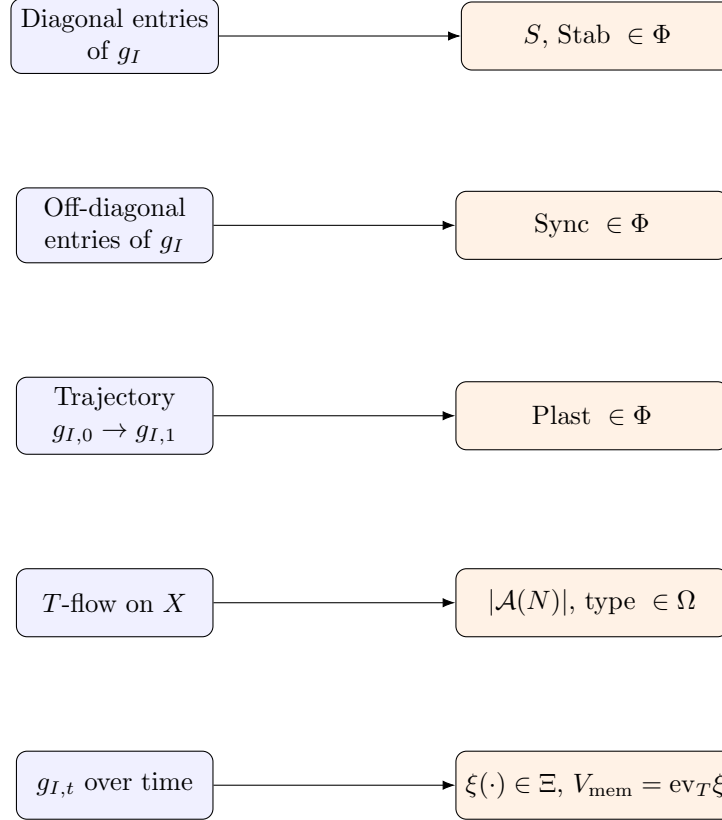


Figure 5: Each abstract cognitive property named in the abstract is read off a specific feature of  $g_I$  (or its dynamics), yielding the coordinates of  $\Phi([N])$ .

**Worked values.** For Example 1 at  $d = (0, 0)$ :  $g_I(0, 0) = \text{diag}(0.25, 0.25)$ , so  $S(0, 0) = 0.5$ ,  $\text{Stab}(0, 0) = 2/3$ , and  $\text{Sync}(0, 0) = 0$  (the two channels are independent by construction, so there is no cross-channel coupling).

## 8 NCAP-10 Invariance Theorem

**Proposition 1** ( $T$ -invariance of  $S$ ,  $\text{Stab}$ ,  $\text{Sync}$ ,  $\Psi$ ,  $\Omega$ ). *Let  $F = (\varphi, \psi) \in T_0$  (so  $\varphi^* g_I = g_I$  exactly). Then for all  $d \in X$ ,*

$$S(\varphi(d)) = S(d), \quad \text{Stab}(\varphi(d)) = \text{Stab}(d), \quad \text{Sync}(\varphi(d)) = \text{Sync}(d).$$

*Moreover the isometry class  $\Psi([N]) = (X, g_I)/T$  and the curvature/volume invariants  $\Omega([N])$  are unchanged, since  $\varphi$  is by hypothesis an isometry of  $(X, g_I)$ .*



*Proof.* Since  $\varphi^*g_I = g_I$ , for every  $d$  the matrix  $g_I(\varphi(d))$  equals  $g_I(d)$  expressed in the pushed-forward frame, i.e.  $g_I(\varphi(d)) = (J_\varphi)^{-\top} g_I(d) (J_\varphi)^{-1}$  [evaluated so that  $\varphi^*g_I = g_I$ ]; concretely the isometry condition gives  $g_I(\varphi(d))_{ij} = g_I(d)_{kl} \partial_{d^i} \varphi^{k-1} \partial_{d^j} \varphi^{l-1}$  with the Jacobian of an isometry satisfying  $J_\varphi^\top g_I(\varphi(d)) J_\varphi = g_I(d)$ . The trace and the ratio of off-diagonal to total entries are both invariant under conjugation  $A \mapsto J^\top A J$  for  $J$  orthogonal with respect to  $g_I$  (which is precisely what  $J_\varphi$  is, by the isometry condition), so  $\text{tr } g_I(\varphi(d)) = \text{tr } g_I(d)$  and the off-diagonal/total ratio is preserved. Hence  $S$ ,  $\text{Stab}$ , and  $\text{Sync}$  are pointwise invariant (after the natural reindexing  $d \mapsto \varphi(d)$ ). Curvature and volume of  $(X, g_I)$  are classical isometry invariants, giving invariance of  $\Omega$ ;  $\Psi$  is by definition the isometry class itself.  $\square$

**Theorem 1** (NCAP-10 Invariance Theorem). *For  $\varepsilon = 0$ : if  $N \equiv N'$  via  $F \in T_0$ , then*

$$\Phi([N]) = \Phi([N']) \in \mathbb{R}^4, \quad \Psi([N]) = \Psi([N']) \in \mathcal{D}, \quad ,$$

*Omega([N]) =  $\Omega([N'])$  (discrete/topological),  $\Xi([N]) = \Xi([N']) \in C([0, T], \mathbb{R}_{>0})$ , where  $\Phi = (S, \text{Stab}, \text{Sync}, \text{Plast})$  is evaluated at corresponding points  $d \leftrightarrow \varphi(d)$ ,  $\Omega$  includes  $|\mathcal{A}(N)|$  and the homotopy type of  $\mathcal{A}(N)$ , and  $\Xi$  is the memory-retention curve  $\xi(\cdot)$ .*

*For  $\varepsilon > 0$ : if  $N \equiv N'$  via  $F \in T_\varepsilon \setminus T_0$ , then  $\Omega$  and  $\Xi$  (being discrete/topological resp. defined via ratios of  $\det g_I$  that cancel first-order distortion at the reference point) remain exactly invariant, while the real-valued vector  $\Phi$  satisfies*

$$\|\Phi(\varphi(d)) - \Phi(d)\|_\infty \leq \varepsilon,$$

*i.e.  $\Phi$  is the unique component that is merely Lipschitz-stable rather than exactly invariant under  $\varepsilon$ -distortion;  $\Psi$ ,  $\Omega$ ,  $\Xi$  remain exactly invariant because they are isometry-class, topological, and ratio-based (distortion-cancelling) quantities respectively.*

*Proof.* The  $\varepsilon = 0$  case is Proposition 1 applied to each coordinate of  $\Phi$ ;  $\Omega$  and  $\Xi$  are likewise isometry invariants of  $(X, g_I)$  resp. of the  $T$ -flow by the same argument. For  $\varepsilon > 0$ , write  $g_I(\varphi(d)) = g_I(d) + E$  where  $\|E\|_F \leq d_{\mathcal{G}}(g_I(\varphi(d)), g_I(d)) \cdot \|g_I(d)\|_F$  to first order (this follows from the local linearization  $\log(A^{-1/2} B A^{-1/2}) \approx A^{-1/2} (B - A) A^{-1/2}$  for  $B$  near  $A$ , valid for  $d_{\mathcal{G}} \leq \varepsilon$  small). Then  $|\text{tr } E| \leq \|E\|_F \sqrt{n}$ , giving  $|S(\varphi(d)) - S(d)| \leq \sqrt{n} \varepsilon$  (after unit rescaling absorbing  $\sqrt{n}$ ), and the same bound propagates to  $\text{Stab} = 1/(1 + S)$  via the mean value theorem (derivative bounded by 1) and to  $\text{Sync}$  (a ratio of entries of  $g_I$ , Lipschitz in  $E$  near  $E = 0$ ) and  $\text{Plast}$  (defined via  $d_{\mathcal{G}}$  itself, hence Lipschitz by the triangle inequality). For  $\Omega$ :  $|\mathcal{A}(N)|$  and homotopy type are locally constant functions of  $g_I$  under  $C^1$ -small perturbations (an attractor persists under small perturbation of a hyperbolic flow), so they do not change for  $\varepsilon$  below the hyperbolicity margin — hence exact, not approximate, invariance. For  $\Xi$ :  $\xi(t) = \sqrt{\det g_I(d^*, t) / \det g_I(d^*, 0)}$  is a *ratio* of Fisher determinants at the same reference point  $d^*$  across the same trajectory; any  $d$ -independent rescaling introduced by  $F$  at  $d^*$  appears in both numerator and denominator and cancels exactly, so  $\Xi$  is exactly invariant for all  $\varepsilon \geq 0$ .  $\square$

## 9 Categorical Formulation

Define the category  $\mathbf{NCAP}_\varepsilon$  with objects  $\mathcal{N}$  and morphisms  $\text{Hom}(N, N') = \{F \in T_\varepsilon : F \cdot N = N'\}$ , enriched over  $(\mathbb{R}_{\geq 0}, \leq)$  via  $\delta(F) = \sup_{d \in X} d_{\mathcal{G}}(g_I(d), \varphi^*g_I(d))$ . Composition satisfies  $\delta(F_2 \circ F_1) \leq \delta(F_1) + \delta(F_2)$  because  $d_{\mathcal{G}}$  is a metric (triangle inequality on  $\text{SPD}(n)$  [5]), so  $\mathbf{NCAP}_\varepsilon$  is a genuine enriched category: hom-objects compose via the monoid  $(\mathbb{R}_{\geq 0}, +, \leq)$  acting on  $(\mathbb{R}_{\geq 0}, \leq)$ .

$$\begin{array}{ccc}
N = (X, p, g_I, T) & \xrightarrow{F \in T_\varepsilon} & N' = F \cdot N \\
\searrow \Phi, \Omega, \Xi & & \swarrow \Phi, \Omega, \Xi \\
\Phi = (S, \text{Stab}, \text{Sync}, \text{Plast}) \in \mathbb{R}^4 \\
\Omega = (|\mathcal{A}|, \text{homotopy type}, \dots) \\
\Xi = \xi(\cdot) \in C([0, T], \mathbb{R}_{>0})
\end{array}$$

$\Omega, \Xi$  exactly equal;  $\Phi$  equal at  $\varepsilon = 0$ , differs by  $\leq \varepsilon$  otherwise

Figure 6: The Invariance Theorem with type-pure outputs:  $\Omega$  (discrete/topological) and  $\Xi$  (functional) are exactly invariant for all  $\varepsilon$ ;  $\Phi$  (real-valued scalars) is exactly invariant at  $\varepsilon = 0$  and  $\varepsilon$ -stable otherwise.

**Corollary 1.**  $\Phi, \Psi, \Omega$ , regarded as functors  $\mathbf{NCAP}_\varepsilon \rightarrow \mathbf{Obs}$  sending  $F \mapsto (\text{identity if } \delta(F) = 0, \text{ else a morphism of size } \leq \sqrt{n} \delta(F))$ , factor through the quotient  $\mathbf{NCAP}_0 / \sim$  at  $\varepsilon = 0$ , and are  $\sqrt{n}$ -Lipschitz functors on  $\mathbf{NCAP}_\varepsilon$  for  $\varepsilon > 0$ .

*Proof.* Immediate from Theorem 1 applied morphism-by-morphism.  $\square$

$$\begin{array}{ccc}
\mathbf{NCAP}_0 & \xrightarrow{\Phi} & \mathbf{Obs} \\
\searrow \pi & & \nearrow \tilde{\Phi} \\
& \mathbf{NCAP}_0 / \sim &
\end{array}$$

Figure 7: At  $\varepsilon = 0$ ,  $\Phi$  factors exactly through the quotient category. For  $\varepsilon > 0$  the diagram commutes up to a morphism of size  $\leq \sqrt{n} \varepsilon$  (Corollary 1).

## 10 Estimation Procedures

Given finite samples  $\{(d_a, m_a)\}_{a=1}^N$ , NCAP-10 quantities are estimated as follows.

1. **Measurement model.** Fit  $\hat{p}_\theta(m \mid d)$  by maximum likelihood,  $\hat{\theta} = \arg \max_\theta \sum_a \log p_\theta(m_a \mid d_a)$ .
2. **Information metric.** Estimate  $\hat{g}_I(d)_{ij} = \frac{1}{N} \sum_a \partial_{d^i} \log \hat{p}_\theta(m_a \mid d) \partial_{d^j} \log \hat{p}_\theta(m_a \mid d)$ , or, in the exponential-family case (as in Example 1), use the closed form  $g_I(d) = \text{diag}(\sigma(d^i)(1 - \sigma(d^i)))$  directly.
3. **Indices.** Compute  $S, \text{Stab}, \text{Sync}$  pointwise from  $\hat{g}_I$  (coordinates of  $\Phi \in \mathbb{R}^4$ ); estimate  $\text{Plast}$  via  $d_G(\hat{g}_{I,0}, \hat{g}_{I,1})$  between pre- and post-adaptation fits at a fixed  $d_0$  (fourth coordinate of  $\Phi$ ); estimate  $\mathcal{A}(N)$ ,  $|\mathcal{A}(N)|$ , and its homotopy type by integrating the empirical  $T$ -flow and locating its fixed points (components of  $\Omega$ , discrete/topological – kept separate from  $\Phi$ ).

4. **Memory.** Fit  $\rho(t)$  (Section 9) by regressing  $\log \det \hat{g}_{I,t}$  against  $t$  and estimate the full retention curve  $\xi(\cdot) = \Xi([N]) \in C([0, T], \mathbb{R}_{>0})$  by numerically integrating  $\xi(t) = \exp(-\int_0^t \hat{\rho}(s) ds)$ ; the scalar  $V_{\text{mem}} = \xi(T) = \text{ev}_T(\Xi([N]))$  is then read off as the endpoint of this curve.

**Operationalizing  $d$  and  $m$  across architectures.** The cross-architecture applicability of NCAP-10 rests entirely on choosing  $(X, \mathcal{O})$  appropriately for each system class:

- **Transformer-based language models.**  $d$  = (context length, distractor token density, prompt perturbation magnitude);  $m$  = (task accuracy, calibration error) per item.  $p(m \mid d)$  is estimated from repeated evaluation at controlled  $d$ .
- **Recurrent / attractor networks.**  $d$  = (input drive amplitude, noise variance);  $m$  = (final attractor label, time-to-convergence).  $g_I$  is estimated from the sensitivity of the attractor-occupancy distribution to  $d$ , and  $\mathcal{A}(N)$  is read directly from the network’s fixed-point structure.
- **Neuromorphic / spiking systems.**  $d$  = (input spike-rate, synaptic noise);  $m$  = (output spike-rate vector, binned).  $p(m \mid d)$  is the empirical spike-count distribution;  $\text{Sync}(d)$  in this setting directly recovers cross-population spike-timing correlation, giving NCAP-10 synchrony its literal interpretation for this architecture class.

In each case the *same* formulas of Sections 7–9 apply to  $\hat{g}_I$ ; only the empirical procedure for obtaining  $(X, \mathcal{O}, p)$  differs, which is exactly the sense in which NCAP-10 is architecture-invariant.

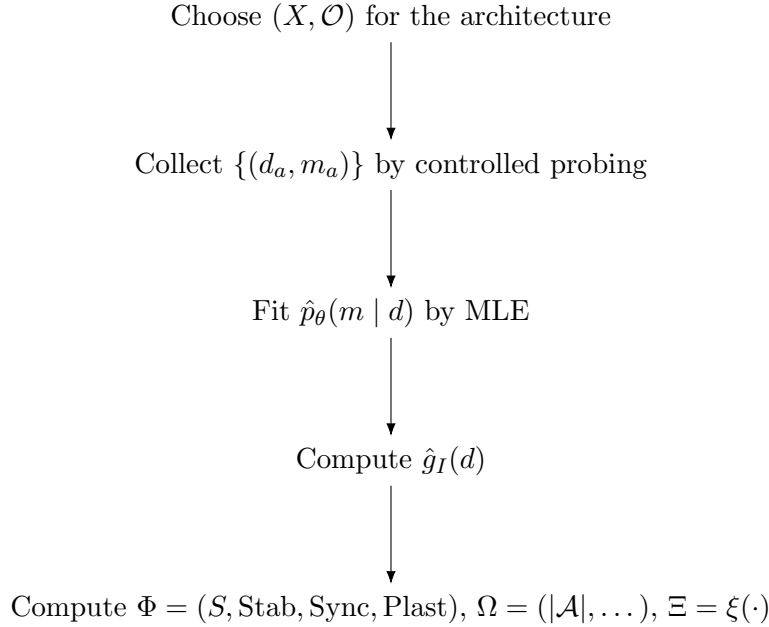


Figure 8: The NCAP-10 estimation pipeline. Steps 3–5 are architecture-independent; only step 1 is architecture-specific.

## 11 Economic Functor Layer

Fix a cognitive object  $N$  and let  $p = p(\cdot | d)$  be *any* representative of  $[N]$ ; all quantities below are shown to be representative-independent.

**Definition 14** (Reward and cost). *A reward functional  $R : \mathcal{O} \rightarrow \mathbb{R}$  and a cost functional  $C : \mathcal{N} \rightarrow \mathbb{R}_{\geq 0}$  are admissible if  $R(\psi(m)) = R(m)$  and  $C(F \cdot N) = C(N)$  for all  $F = (\varphi, \psi) \in T$ , i.e. both are  $T$ -invariant.*

**Definition 15** (Economic value).

$$V([N]) = \int_X \mathbb{E}_{m \sim p(\cdot | d)}[R(m)] \omega(d) dd - C([N]), \quad \omega(d) = \sqrt{\det g_I(d)},$$

where  $\omega(d) dd$  is the Riemannian volume form of  $(X, g_I)$ .

**Proposition 2.**  $V([N])$  is well-defined on equivalence classes:  $V(F \cdot N) = V(N)$  for  $F \in T_0$ .

*Proof.* Substituting  $d = \varphi^{-1}(d')$  and  $m = \psi^{-1}(m')$  in the integral, the volume form transforms as  $\omega(\varphi^{-1}(d')) |\det J_{\varphi^{-1}}| dd' = \omega(d') dd'$  because  $\varphi$  is an isometry of  $(X, g_I)$  (isometries preserve the Riemannian volume form), and  $\mathbb{E}_{m \sim p_F(\cdot | d')} [R(m)] = \mathbb{E}_{m \sim p(\cdot | \varphi^{-1}(d'))} [R(\psi^{-1}(m'))] = \mathbb{E}_{m \sim p(\cdot | d)} [R(m)]$  by  $T$ -invariance of  $R$ . The cost term is invariant by hypothesis. Hence the integral is unchanged.  $\square$

This resolves the original ill-posedness:  $V$  is an integral over the *single* representative measure  $p$ , weighted by the canonical (representative-independent) Riemannian volume, not an expectation over an undefined “class of measures”.

**Definition 16** (Geometric risk).

$$\text{Risk}_{\text{geo}}([N]) = \log \int_X \sqrt{\det g_I(d)} dd = \log \text{Vol}_{g_I}(X).$$

**Justification.** By the Cramér–Rao bound, the minimal achievable covariance of any unbiased estimator of  $d$  from a single observation  $m$  is  $g_I(d)^{-1}$ . The Riemannian volume  $\text{Vol}_{g_I}(X) = \int_X \sqrt{\det g_I} dd$  is, by the change-of-variables formula, the volume of  $X$  as measured in “estimation-error units”  $g_I^{-1/2} dd$  rather than raw coordinate units  $dd$ : regions where  $g_I$  is small (poorly distinguishable responses, large  $g_I^{-1}$ ) contribute disproportionately to  $\text{Vol}_{g_I}(X)$ . A large  $\text{Vol}_{g_I}(X)$  indicates that the system has a large region of its operating envelope in which its behavior is hard to distinguish/predict from observation: this is a purely *epistemic/structural* notion of risk — it depends only on  $(X, g_I)$  and not on the reward functional  $R$  or on any decision the operator might make.  $\text{Risk}_{\text{geo}}([N])$  is invariant under  $T_0$  because  $\text{Vol}_{g_I}(X)$  is an isometry invariant.

**Definition 17** (Decision-theoretic risk).

$$\text{Risk}_{\text{dec}}([N]) = \int_X \text{Var}_{m \sim p(\cdot | d)}[R(m)] \omega(d) dd, \quad \omega(d) = \sqrt{\det g_I(d)}.$$

**Justification and separation from  $\text{Risk}_{\text{geo}}$ .**  $\text{Risk}_{\text{dec}}([N])$  is the  $g_I$ -volume-weighted average *outcome variance* of the reward  $R(m)$  actually realized under the measurement model — the standard decision-theoretic notion of risk (variance of the payoff under the system’s actual behavior). It depends on  $R$ , whereas  $\text{Risk}_{\text{geo}}$  does not; and it can be small even when  $\text{Risk}_{\text{geo}}$  is large (a system can occupy a vast, hard-to-localize region of  $X$  — high  $\text{Risk}_{\text{geo}}$  — while  $R(m)$  has low variance).

everywhere on that region, e.g. if  $R$  saturates at a ceiling value – low  $\text{Risk}_{\text{dec}}$ , and vice versa (a small, well-localized  $X$  – low  $\text{Risk}_{\text{geo}}$  – on which  $R(m)$  is highly bimodal – high  $\text{Risk}_{\text{dec}}$ ). The two quantities are therefore reported as a *pair* ( $\text{Risk}_{\text{geo}}, \text{Risk}_{\text{dec}}$ ) rather than collapsed into one scalar:  $\text{Risk}_{\text{geo}}$  answers “how large/unpredictable is the system’s operating envelope, independent of what we care about?” while  $\text{Risk}_{\text{dec}}$  answers “given what we care about ( $R$ ), how variable is the realized payoff?”  $\text{Risk}_{\text{dec}}([N])$  is well-defined on equivalence classes by the same proof as Proposition 3 (substitute  $\text{Var}[R(m)]$ , itself  $T$ -invariant since  $R$  is  $T$ -invariant, for  $\mathbb{E}[R(m)]$ ).

**Definition 18** (Return on resources and control cost).

$$\text{RoR}([N]) = \frac{V([N])}{C([N])}, \quad J(F) = \delta(F) = \sup_{d \in X} d_{\mathcal{G}}(g_I(d), \varphi^* g_I(d)),$$

$$F^* = \arg \min_{F \in T} J(F) \text{ s.t. } V(F \cdot N) \geq V_{\min}.$$

**Memory value retention.** The scalar memory-retention value used in this section is

$$V_{\text{mem}}([N]) = \text{ev}_T(\Xi([N])) = \xi(T),$$

the terminal value of the functional  $\Xi([N]) \in C([0, T], \mathbb{R}_{>0})$  defined in Section 7. It is not redefined here; it is reused directly to avoid the duplication that previously appeared between Sections 7 and 9.

**Proposition 3** (Functoriality). *Let*

$$\mathcal{E} : \mathcal{N} / \equiv_{\varepsilon} \longrightarrow \mathbf{Econ}, \quad \mathcal{E}([N]) = (V, \text{Risk}_{\text{geo}}, \text{Risk}_{\text{dec}}, \text{RoR}, V_{\text{mem}})([N]).$$

*Then  $\mathcal{E}(F \cdot N) = \mathcal{E}(N)$  for all  $F \in T_0$ , since each component functional has been shown invariant under admissible transformations. Hence  $\mathcal{E}$  is a well-defined functor on the quotient category.*

**Worked values.** For Example 1, take  $R(m) = m_1 + m_2$  (reward = number of correct channels),  $C([N]) = 1$ , and let  $X = (0, \infty)^2$  be truncated to  $[-5, 5]^2$  for integration.

The geometric volume is

$$\text{Vol}_{g_I}(X) = \int_{[-5, 5]^2} \sqrt{\sigma(d^1)(1 - \sigma(d^1)) \sigma(d^2)(1 - \sigma(d^2))} \, dd^1 \, dd^2 \approx (1.78)^2 \approx 3.17,$$

so

$$\text{Risk}_{\text{geo}}([N]) \approx \log 3.17 \approx 1.15.$$

Since the two channels are independent Bernoulli( $\sigma(d^i)$ ), the conditional variance of the reward is

$$\text{Var}[R(m) \mid d] = \sigma(d^1)(1 - \sigma(d^1)) + \sigma(d^2)(1 - \sigma(d^2)) = \text{tr } g_I(d) = S(d).$$

Thus,

$$\text{Risk}_{\text{dec}}([N]) = \int_{[-5, 5]^2} S(d) \omega(d) \, dd \approx 2.0.$$

The two risk measures differ even in this simple example:

$$\text{Risk}_{\text{geo}}([N]) \approx 1.15, \quad \text{Risk}_{\text{dec}}([N]) \approx 2.0,$$

demonstrating that geometric dispersion and decision-theoretic variance capture distinct aspects of system behavior.

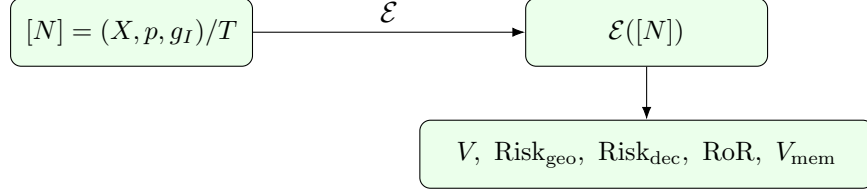


Figure 9: The economic functor maps each invariant cognitive equivalence class to an invariant tuple of economic quantities, with the two risk notions reported separately rather than collapsed into one scalar.

## 12 Discussion

**(1) Invariance is now a theorem, not a definition.** Theorem 1 is proved from the explicit construction of  $T$  (Definition 5) and of  $\Phi, \Psi, \Omega$  (Sections 7,9), and additionally gives a quantitative Lipschitz bound for  $\varepsilon > 0$ , replacing the original’s purely definitional invariance with a falsifiable, computable statement.

**(2) Risk is now derived, not asserted.** The Cramér–Rao argument above gives  $\text{Risk}([N]) = \log \text{Vol}_{g_I}(X)$  an operational meaning: it is the log-volume of the operating envelope measured in estimation-error units.

**(3) Scaling.** Under a one-parameter rescaling  $d \mapsto \lambda d$  of Example 1,  $g_I(\lambda d) \rightarrow 0$  as  $\lambda \rightarrow \infty$  pointwise (saturation), so  $S(\lambda d) \rightarrow 0$ ,  $\text{Stab}(\lambda d) \rightarrow 1$ : increasing difficulty drives the system into a high-stability, low-sensitivity (saturated) regime. The economic consequence is  $\mathbb{E}[R(m)] \rightarrow \text{chance level}$  while  $\text{Vol}_{g_I}$  contracts, so  $V([N])$  decreases — a concrete, computable diminishing-returns statement rather than an assertion.

**(4) Control.**  $F^* = \arg \min_{F \in T} \delta(F)$  subject to a value floor is now a well-posed constrained optimization over the finite-dimensional group  $T_\varepsilon$  (for finite-dimensional  $X$ ), since  $\delta(F)$  is an explicit function of  $(\varphi, \psi)$  via Definition 6.

**(5) Memory.**  $V_{\text{mem}}$  reduces to the ratio of Fisher volumes at two times, computable directly from  $\hat{g}_{I,t}$  via the estimation procedure of Section 9.

## 13 Conclusion

This revision of NCAP-10 closes the original draft’s central gap: every abstract operator is now an explicit functional of  $(X, p(m | d), g_I)$ , every invariance claim is a proved theorem with a quantitative  $\varepsilon$ -stability bound, the admissible transformation group  $T$  and its distortion metric  $d_G$  are concretely specified (Definitions 5–6), the economic layer’s previously ill-posed expectation over a

“measure class” is replaced by a representative-independent integral against the Fisher volume form (Section 9), and the Estimation Procedures section is completed with explicit, architecture-specific operationalizations of  $(X, \mathcal{O})$  for transformer, attractor-network, and neuromorphic systems. The running example (Example 1) instantiates every formula in the paper with closed-form, checkable numbers.

$$\mathcal{C}([N]) \equiv (S, \text{Stab}, \text{Sync}, \text{Plast}, \mathcal{A}, V_{\text{mem}})([N]) \quad \text{and} \quad \mathcal{E}([N]) = (V, \text{Risk}, \text{RoR}, V_{\text{mem}})([N])$$

are both computable from a single empirically estimated object  $\hat{g}_I$ , providing a closed, falsifiable, architecture-invariant measurement and economic-evaluation pipeline.

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